

Code Review Report

EXT003MI / CompleteTransfr

Review focus: database access, performance, maintainability, transaction flow, and XtendM3 approval readiness.

Excluded from this report as requested: error handling, type-safe comparison, update correctness, and logging review.

Extension Name	EXT003MI / CompleteTransfr
Extension Type	ExtendM3Transaction
Version Reviewed	2.0
Revision Note	SORT and ITNO removed from readAll and changed to read
Reviewer	Microsoft 365 Copilot

1. Executive Summary

The extension is organized well and the latest revision improves the SORT and ITNO lookup pattern by using single-record database reads where the key values are available. The code has a clear transaction flow: read delivery data, derive deck and sort details, call the deck transfer MI, update deck detail balances, and return MI output.

Based on the requested review scope, the main remaining recommendations are around database access optimization, reducing unnecessary readAll usage, improving maintainability, strengthening transaction design clarity, and making the code easier to support during future changes.

2. Review Scope

Reviewed Area	Status	Remarks
Database access pattern	Reviewed	Focused on read vs readAll usage, index usage, and selection fields.
Performance	Reviewed	Checked whether single-record lookups are used when the key is known.
Maintainability	Reviewed	Checked method structure, variable usage, naming consistency, constants, and documentation.
Transaction flow	Reviewed	Checked whether the sequence is clear and easy to support.
Error handling	Excluded	Excluded based on user instruction.
Type-safe comparison	Excluded	Excluded based on user instruction.
Update correctness	Excluded	Excluded based on user instruction.
Logging	Excluded	Excluded based on user instruction.

3. Positive Observations

Observation	Details
Logical method separation	The code separates table lookups and processing into individual methods such as findEXTDLH, findEXTDTV, findEXTDPH, findEXTSOR, findEXTDPD, getSunm, addDeckTransferMI, and updEXTDPDRecord. This improves readability and supportability.
Good use of selection fields	Most database actions use selection fields, which avoids reading unnecessary columns and makes intent clearer.
Improved SORT and ITNO lookup	The current revision uses read for EXTDPH and EXTSOR, which is appropriate when the full key is available and a single record is expected.
Header comments and revision history	The script includes extension metadata and revision history, which is helpful for approval review and future maintenance.
Separate MI call wrapper	The EXT005MI.AddDeckTransfer call is isolated in a dedicated method, making the integration point easier to identify and modify.

4. Findings and Recommendations

F1. CIDMAS lookup should use read instead of readAll

Priority: Medium

Observation: In `getSunm()`, the code uses `readAll` even though company and supplier number are available. If CIDMAS index 00 is keyed by CONO and SUNO, a single read is more suitable and avoids unnecessary callback iteration.

Recommendation: Change `getSunm()` to use `queryCIDMAS.read(CIDMAS)`. This improves performance and makes the code intent clearer.

F2. Review EXTCPI readAll usage

Priority: Medium

Observation: `findEXTCPI()` uses `readAll` and stores EXPINO into a single variable. If multiple payment records are returned, only the final callback value remains in `inPino`. This may be intentional, but the rule is not visible in the code.

Recommendation: If one payment record is expected, use `read`. If multiple records are valid, document which record should be selected and why.

F3. pageSize value is high for single-record style reads

Priority: Low

Observation: `pageSize` is set to 10000. The value is only required for `readAll` operations. Using such a large page size can be misleading when most lookups are single-record reads.

Recommendation: Keep `pageSize` only for genuine list reads. For single-record database access, prefer `read` and remove dependency on `pageSize`.

F4. Class-level variables make transaction state harder to track

Priority: Medium

Observation: Many working variables are declared at class level. This makes the transaction flow harder to reason about because values are assigned across multiple methods.

Recommendation: Use local variables where possible. Keep class-level variables only when values must be shared with callbacks or multiple methods.

F5. Consider constants for MI program, transaction, and table names

Priority: Low

Observation: String literals such as `EXT005MI`, `AddDeckTransfer`, `EXTDLH`, `EXTDTV`, `EXTDPH`, `EXTSOR`, `EXTDPD`, `EXTCPI`, and `CIDMAS` are repeated directly in methods.

Recommendation: Define constants for repeated program names, transaction names, table names, and index names. This reduces typo risk and improves maintainability.

F6. Method naming can be refined for clarity

Priority: Low

Observation: Methods named `findEXTDLH`, `findEXTDTV`, and `findEXTDPD` are clear for technical readers, but business intent is not visible in the method name.

Recommendation: Consider names such as `findDeliveryHeader`, `findDeliveryVolume`, `findDeckHeader`, `findSort`, and `findDeckDetail` if project conventions allow business-oriented names.

F7. Header description can be more specific

Priority: Low

Observation: The header says the extension is used to create a distribution order in M3, while the logic appears focused on completing transfer flow and calling AddDeckTransfer.

Recommendation: Update the script description to reflect the actual transaction purpose: completing transfer, creating deck transfer, and updating deck detail balance.

F8. Output field mapping should be documented

Priority: Low

Observation: The extension returns DLNO, ITNO, WBAL, ABAL, PINO, and TRNO. The source and meaning of these outputs are not documented near the response block.

Recommendation: Add a short comment before mi.outData.put calls to document each output field. This helps MI consumers and future reviewers.

5. Suggested Code Improvements

The following examples are limited to the reviewed scope and do not include the excluded areas.

Suggested getSunm() using read

```
void getSunm(){
    DBAction queryCIDMAS = database.table("CIDMAS").index("00").selection("IDSUNM").build()
    DBContainer CIDMAS = queryCIDMAS.getContainer()
    CIDMAS.set("IDCONO", CONO)
    CIDMAS.set("IDSUNO", inSUNO)

    if(queryCIDMAS.read(CIDMAS)){
        inSUNM = CIDMAS.getString("IDSUNM").trim()
    }
}
```

Suggested constants

```
private static final String MI_PROGRAM_EXT005MI = "EXT005MI"
private static final String MI_TRANSACTION_ADD_DECK_TRANSFER = "AddDeckTransfer"
private static final String TABLE_CIDMAS = "CIDMAS"
private static final String INDEX_00 = "00"
```

6. Approval Readiness Checklist

Checklist Item	Result	Comment
Uses read when single-record key is available	Partial	EXTDPH and EXTSOR are good. CIDMAS and possibly EXTCPI can be improved.
Uses database selection fields	Pass	Selection fields are used in most database reads.
Method structure is readable	Pass	Table access and MI call logic are separated into dedicated methods.
Avoids unnecessary callback iteration	Needs improvement	CIDMAS and EXTCPI should be reviewed for readAll replacement or documented list-read intent.
Revision history is maintained	Pass	Version 2.0 note is available in the header.
Business purpose is clearly documented	Needs improvement	Header description should be aligned with the actual transfer completion flow.

7. Final Recommendation

The code is generally well structured and the revision to use read for SORT and ITNO is a positive improvement. Within the requested scope, the main changes recommended before approval are to replace unnecessary readAll calls where a single record is expected, review EXTCPI selection behavior, reduce reliance on broad class-level state, define constants for repeated literals, and align the script header description with the actual transaction behavior.

Review conclusion: Minor to medium improvements recommended before submission, mainly for performance clarity and maintainability.